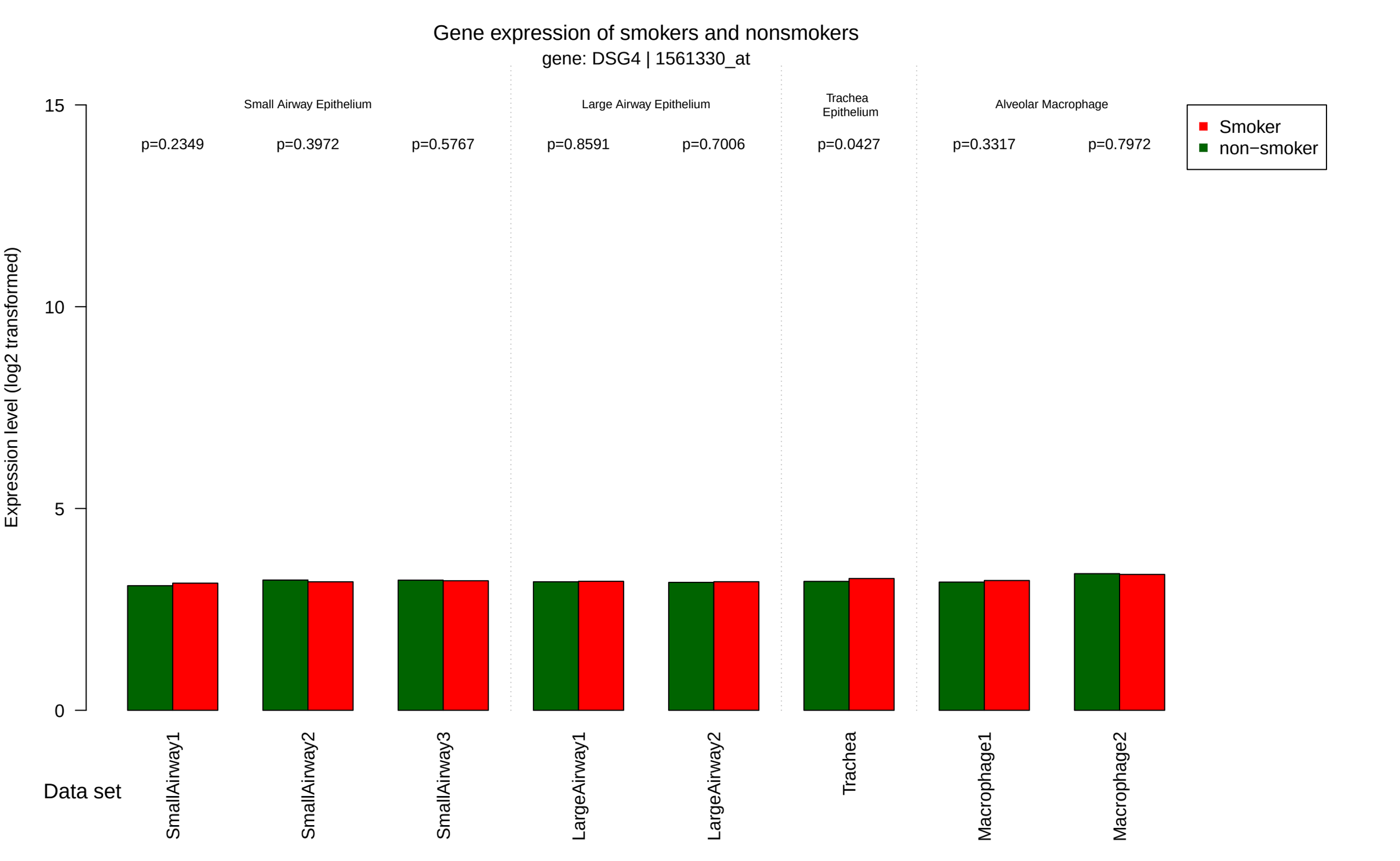
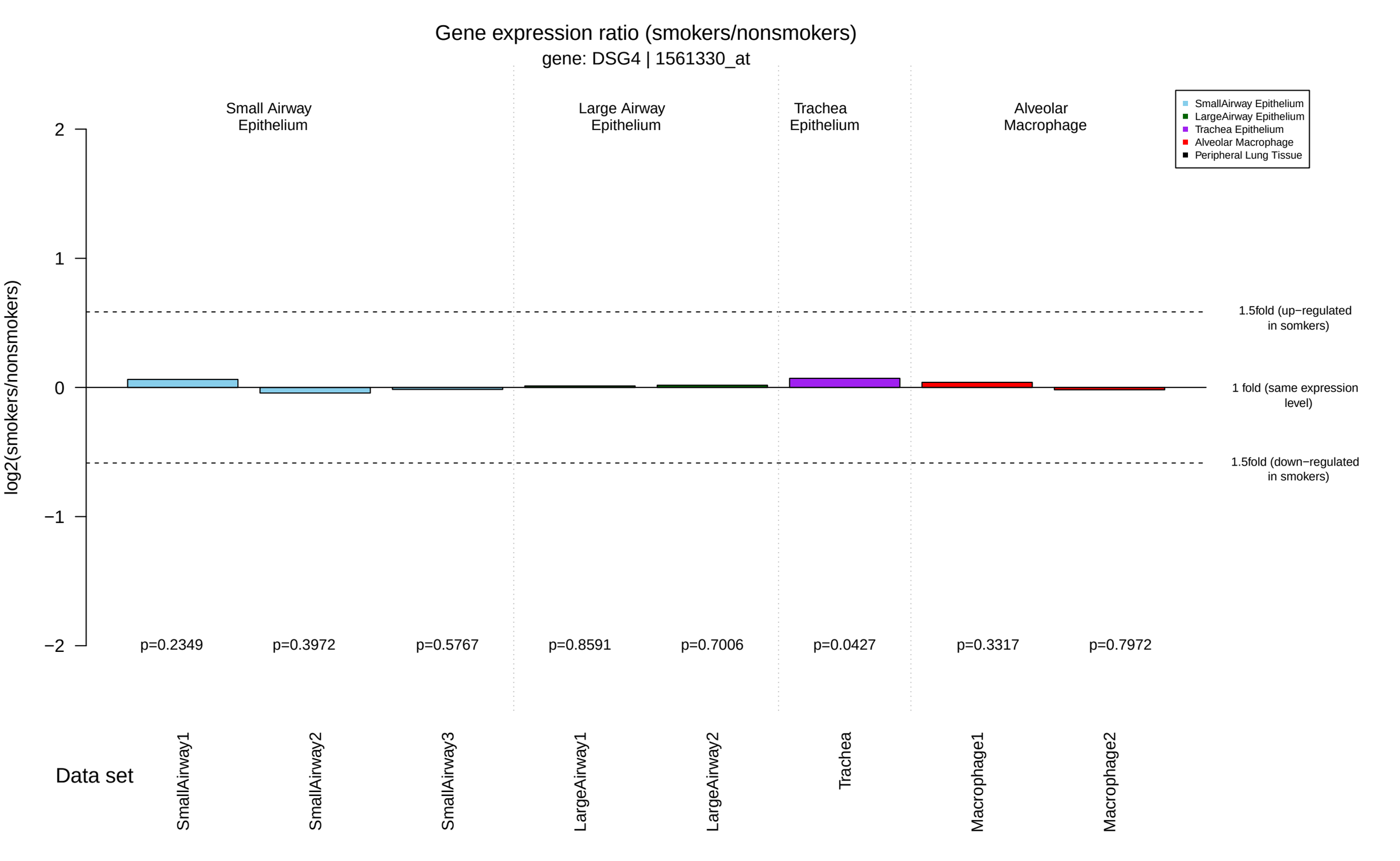
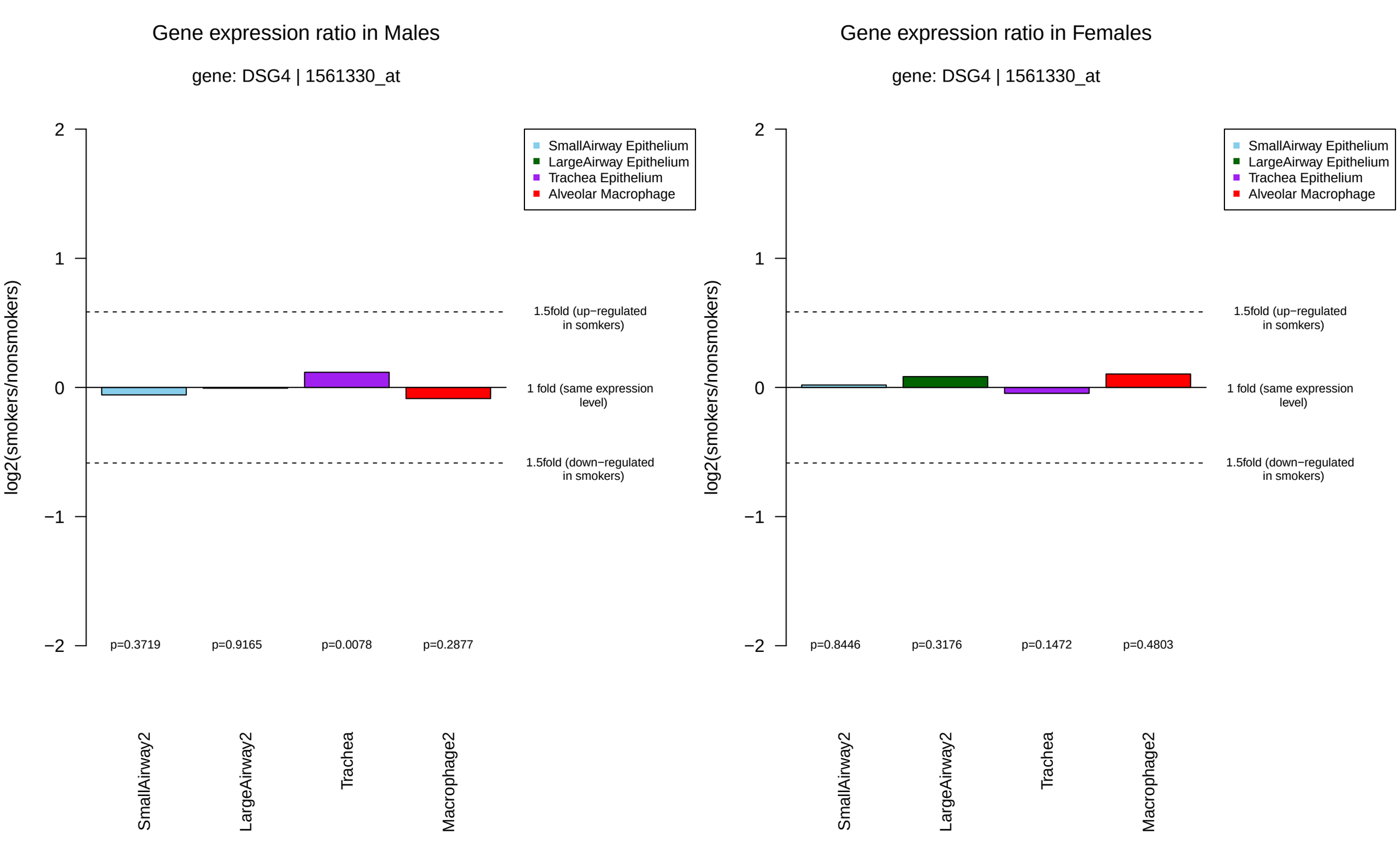
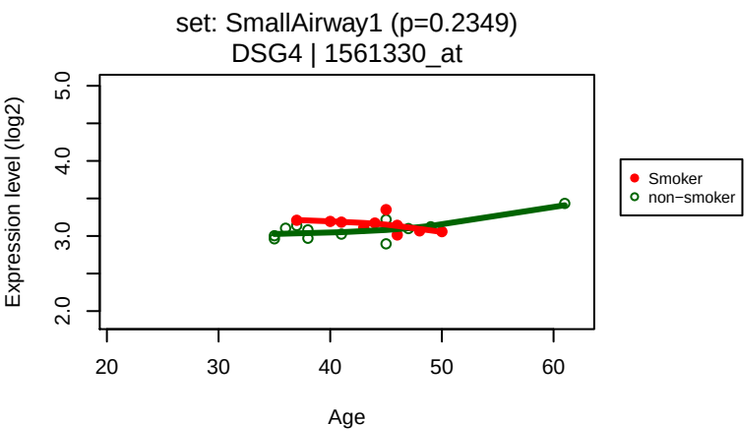
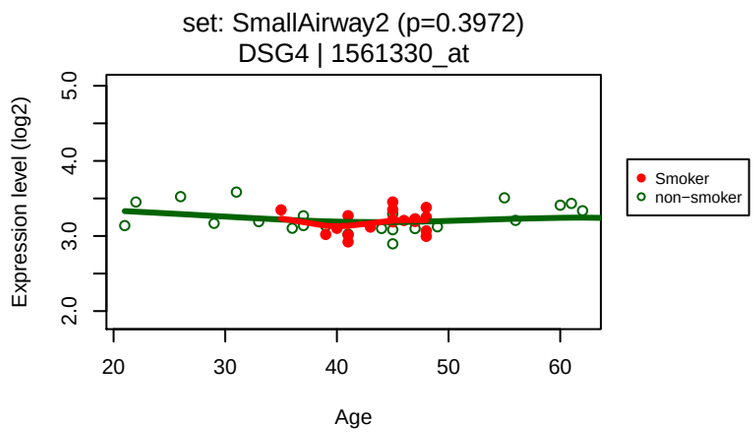
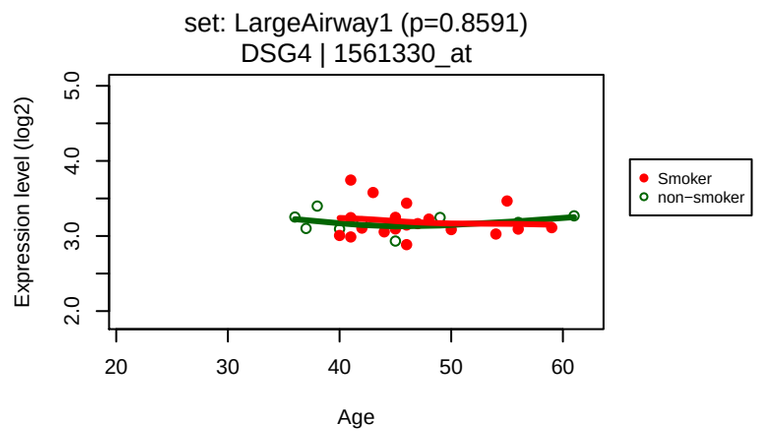

Gene expression of smokers and nonsmokers
gene: DSG4 | 1561330_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2349
p=0.3972
p=0.5767
p=0.8591
p=0.7006
p=0.0427
p=0.3317
p=0.7972
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

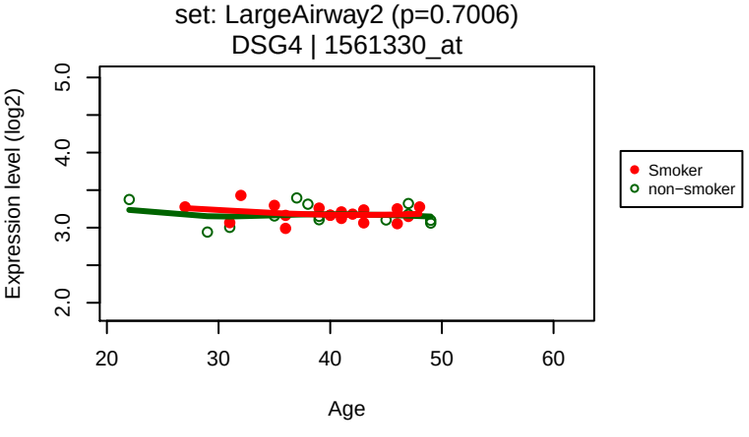

Gene expression ratio (smokers/nonsmokers)
gene: DSG4 | 1561330_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2349
p=0.3972
p=0.5767
p=0.8591
p=0.7006
p=0.0427
p=0.3317
p=0.7972
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

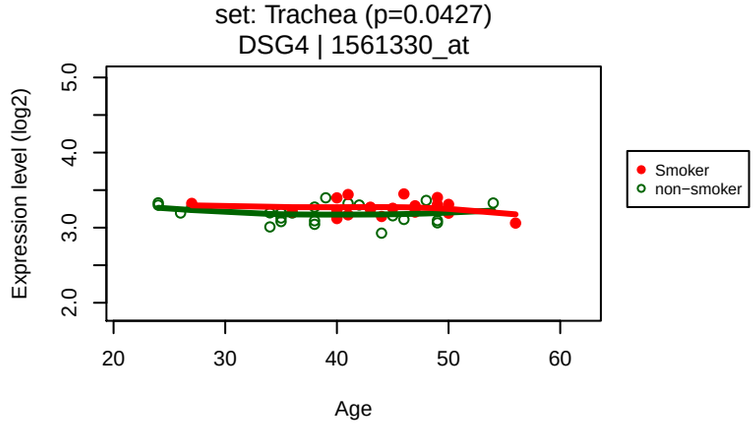

Gene expression ratio in Males
gene: DSG4 | 1561330_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3719
p=0.9165
p=0.0078
p=0.2877
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: DSG4 | 1561330_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8446
p=0.3176
p=0.1472
p=0.4803
SmallAirway2
LargeAirway2
Trachea
Macrophage2

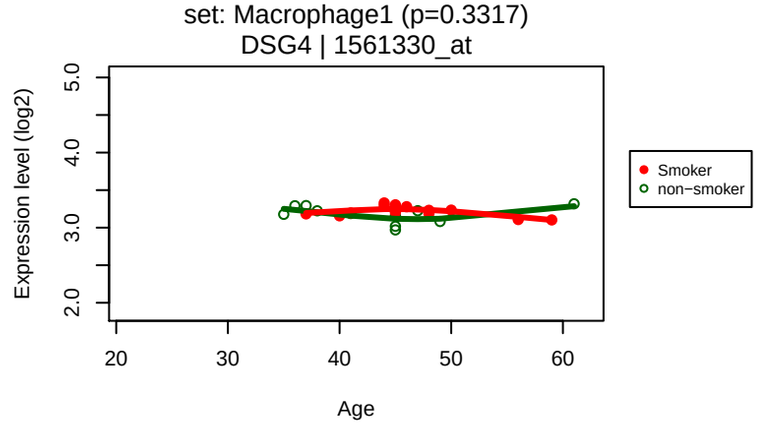

set: SmallAirway1 (p=0.2349)
DSG4 | 1561330_at
Expression level (log2)
Age
Smoker
non−smoker

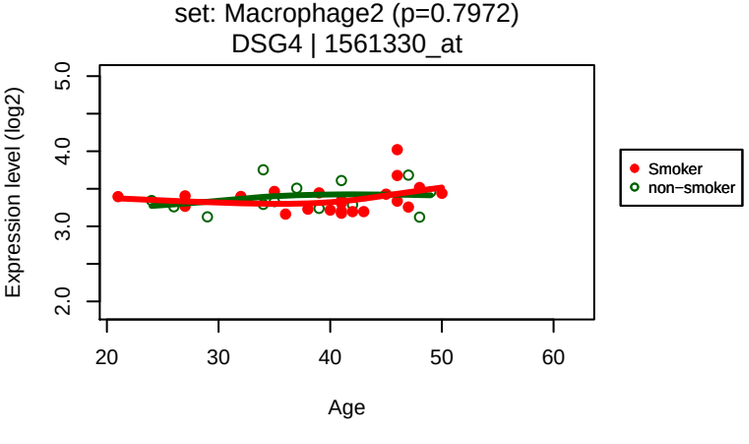

set: SmallAirway2 (p=0.3972)
DSG4 | 1561330_at
Expression level (log2)
Age
Smoker
non−smoker

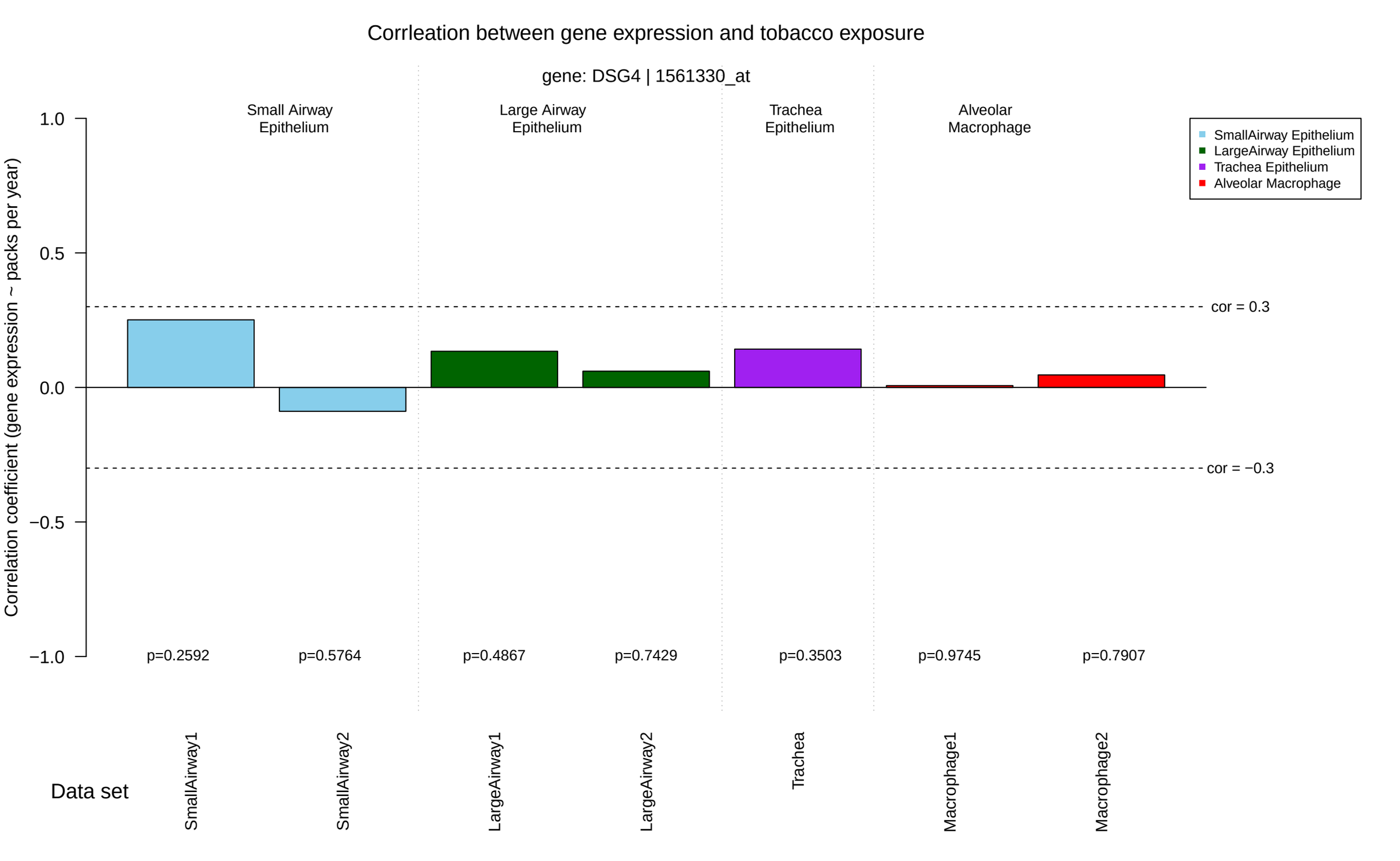

set: LargeAirway1 (p=0.8591)
DSG4 | 1561330_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7006)
DSG4 | 1561330_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0427)
DSG4 | 1561330_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3317)
DSG4 | 1561330_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7972)
DSG4 | 1561330_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: DSG4 | 1561330_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2592
p=0.5764
p=0.4867
p=0.7429
p=0.3503
p=0.9745
p=0.7907
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2